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1) a) $Q = 2000(1000 - P) = 2{,}000{,}000 - 2000P$

$\Rightarrow P = 1000 - Q/2000$

Revenue $= 1000q - q^2/2000$

$MR = \frac{d}{dq}(\text{Revenue}) = 1000 - q/1000$

$MR = MC$

$1000 - q/1000 = 200 \Rightarrow 800 = q/1000$

$q = 800K$

Since, $P = 1000 - q/2000$

$P = 1000 - \frac{800{,}000}{2000}$

$P = \$600$ ← Maximize Profit Price

Contribution to R&D & fixed cost

= Revenue − Variable cost

$= (P - MC)Q = (600 - 200)\,800k$

$= \$320M$

b) No, price does not depend on past R&D, also it does not depend on future R&D. Both are fixed costs with respect to current iPad Sales, and only marginal costs enter the pricing decision.

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c) $Q = 2200(1000 - P)$

$P = 1000 - Q/2200$

$Rev = 1000Q - Q^2/2200$

$MR = 1000 - Q/1100$

$MR = MC$

$1000 - Q/1100 = 200 \Rightarrow 800 = Q/1100$

$Q = 880{,}000$

$P = 1000 - 880{,}000/2200$

$= 1000 - 400$

$= \$600$

Quantity goes up from 600k to 880k 10% ~~increase~~ increase.

Ans: No, apple should not charge a different price ~~it~~ - it stays at $600, because expanding number of customers at each price doesn't change the maximiy Profit Markup price.

d) Before Campaign

Contribution Margin: $(P-MC)Q = 400 \times 800k$

~~$3.2M~~ $320M

After Campaign

Contribution Margin $= 400 \times 880k$

$= \$352M$

Most Apple will pay: $\$352M - \$320M$

$= \$32M$

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e) $Q = 2000(1100 - P)$

$P = 1100 - Q/2000$

Revenue $= 1100Q - Q^2/2000$

$MR = 1100 - Q/1000$

$1100 - Q/1000 = 200$

$900 = Q/1000 \Rightarrow Q = 900{,}000$

$P = 1100 - 450{,}000/2000 = \$650$

Yes, Apple should raise its price to $650, The campaign increases willingness to pay ie increase intercept from 1000 to 1100 rather than expanding customer base, so the optimal markup over marginal cost increases too.

f) After campaign

Contribution Margin $= (650 - 200)\,900{,}000$

$= 450 \times 900k$

$= \$405M$

Incremental profit $= 405 - 320$

$= \$85M$

2a) No, ZEV should not affect Tesla price because ZEV credit is a seperate revenue stream- they dont affect the marginal cost of producing Model S & doesn't affect the demand for the cars. Optimal Pricing $MR = MC$, only depends on the cars Marginal cost & demand curve.

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2b) No, lower interest expense should not affect the Model S price, since it is a fixed cost - it doesn't vary with how many cars Tesla produces, so it's not part of marginal cost & it doesn't shift the demand curve of the car.

c) No, this doesn't change my answer to a or b. Even under a capacity constraint, price is set to clear demand at the fixed quantity capacity, not by marginal cost considerations. Since, neither ZEV credit revenue or financing costs shifts the demand curve, they remain ~~irrel~~ irrelevant to the optimal price.

Ans 3

a) $e_total = (Q_mil / Q_total) \times |e_mil| + (Q_stu / Q_total) \times |e_stu|$

$= .1 \times (2) + .9 (1.2) = 1.1$

$MR = P\left(1 + \frac{1}{e_total}\right)$

$= 25\left(1 - \frac{1}{1.1}\right)$ (Negative, since Mentioned "Absolute Value")

$= \frac{25}{11} \sim \$2.27$

Since, MR > MC

Raising price will decrease Profit

Rather they should decrease cost to increase profit

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b) "$2M liability for every million dose sold" means $2 per dose increase in MC

So, MC = $3 /dose

MR = $2.27 (unchanged)

MR < MC

Increase price would increase profit

c) Risk aversion reinforces the conclusion from (b) - raising prices clearly is more profitable

Ans 4

a) Pac-12

Q = 150k - 3000P

P = 50 - Q/3000

~~Revenue = PQ~~

MR = 50 - Q/1500

MC = 0

Since, MC is offset by profit margin of food & drinks sold.

MR = MC

50 - Q/1500 = 0 ⇒ Q = 75,000

Capacity is 60,000 ← Q

P = 50 - 60000/3000

P = $25

Non-Conference: Q = 90K - 3000P

P = 30 - Q/3000

Q = 30 × 1500 = 45000 ← Q

P = 30 - 45k/3k = 30 - 15

P = $15

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Big Game

$Q = 240k - 3000P$

$P = 80 - Q/3000$

$Q = 80 \times 1500$

$= 120{,}000$

$Q = 6000$ (since capacity is 60k)

$P = \cancel{240000}\ 80 - 60000/3000$

$= 80 - 20$

$P = \$60$

b) MR at $Q = 60000$ for each capacity bound game

Pac 12: $MR = 50 - 2(60000/3000) = 50 - 40 = \$10$

Big-game: $MR = 80 - 2(60000/3000) = 80 - 40 = \$40$

Non-conference $= \$0$

Total value of extra seat

$= 3 \times 0 + 3 \times \$10 + 1 \times \$40$

$= 30 + 40 = \$70$

Since cost is \$75

net loss is $75 - 70 = \$5$

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5 Price-Bots

a) $q = 50 - P$

4000 people in Market A

$Q_a = 4000 \times (50 - P) = 200{,}000 - 4000P$

$P = 50 - Q_a/4000$

$MR = 50 - Q_a/2000$

$MC = \frac{d}{dQ}(5000 + 20Q) = 20$

$50 - Q_a/2000 = 20 \Rightarrow Q_a = 30 \times 2000$

$\boxed{Q_a = 60{,}000}$ $\boxed{P = \$35}$

b) $q = 120 - 2P$

1200 people in Market B

$Q_b = 144{,}000 - 2400P$

$P = 60 - Q_b/2400$

$60 - 2Q_b/2400 = 20 \Rightarrow Q_b = 40 \times 1200$

$\boxed{Q_b = 48{,}000}$ $\boxed{P = \$40}$

$Profit_A = (35 - 20) \times 60k$

$= \$900{,}000$

$Profit_B = (40 - 20) \times 48k$

$= \$960{,}000$

Total profit $= 900k + 960k - 5000$

$= \$1{,}855{,}000$

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c) $Q_a = 200k - 4000P$ when $P \leq 50$
$Q_b = 144k - 2400P$ when $P \leq 60$

for $P \leq 50$, when both are buying
$Q_T = Q_a + Q_b$
$= 344k - 6400P$
$P = 53.75 - Q/6400$
$53.75 - Q/3200 = 20 \Rightarrow Q = 33.75 \times 3200$
$Q = 108,000$
$P = \$36.875$

Profit $= (P-20)Q - 5000$
$= \$1,817,500$

Compared to b Profit \$1,855,000, loss of \$37,500
Since, arbitrage caught up with the markets, we need to compromise with a loss of \$37,500.

6 Golf

a) Aggregated demand
$Q = 100,000 - 10000P$
$P = 10 - Q/10000$
$10 - 2Q/10000 = 2 \Rightarrow Q = 8 \times 10000/2 = 40,000$
$Q = 40,000$ $P = \$6$

Profit $= ((6-2) \times 40000) - 150,000$
$= 160,000 - 150,000$
$= \$10,000$

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b) Choke price ie price where demand hits 0
$q = 0 \quad P = \$10$

Height = Choke price - actual price
= \$10 - \$2 = \$8

Area = $\frac{1}{2}$ × quantity purchased × height

(consumer surplus) = ~~Quantity at Price \$2~~

Quantity at \$2
= 100000 - 20000
= 80000

Quantity at Price = \$2
Q = 100,000 - 2 × 100% = 80000 games each / month
↑ q

Consumer Surplus = $\frac{1}{2}$ × 8 × (10 → 2) = 32

Membership fee = \$32 / month
Revenue from membership = ~~\$320000~~ \$320,000

Profit = 320,000 - 150000
= ~~\$170~~ \$170,000

The per game price is set to \$2 (=MC)

I. Carrot Juice

a) $q = 1000 - 50p$
$p = 20 - q/50$
Revenue = $20q - q^2/50$
$MR = 20 - \frac{q}{25}$

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b) $MC = dC/dQ = 10$

$MR = MC$

$\Rightarrow 20 - 2q/50 = 10 \Rightarrow q = 10 \times 50/2$

$q = 250$ & $P = 20 - 250/50$

$P = \$15$

$Profit = ((15-10) \times 250) - 1000 = 1250 - 100$

$= \$250$

c) $Q_T = 1500 - 50P_T$

$P_T = 30 - Q_T/50$

$30 - q_T/25 = 10 \Rightarrow q_T = 20 \times 25$

$= 500$

$q_T = 500$ & $P_T = 30 - 10 = \$20$

Cal discount = \$5

Effect of tourist price on student price

No, effect. student still pay \$15, same as (b)

Two seperate priced markets

d) Capacity constraint: $Q_S + Q_T = 550$

$MR1 = MR2 \Rightarrow 20 - Q_S/25 = 30 - Q_T/25$

$\Rightarrow Q_T - Q_S = 10 \times 25 = 250$

$Q_T = 400$ & $Q_S = 150$

$P_T = 30 - 400/50 = \$22$ (Standard Price)

$P_S = 20 - 150/50 = \$17$

Cal discount = \$22 - \$17 = \$5

Effect on students: Yes, there is effect due to capacity constraint of 550 drivers & tourists willing to pay higher.